

ScreenConcealer: Privacy-protection System with Obfuscations for Screen Sharing

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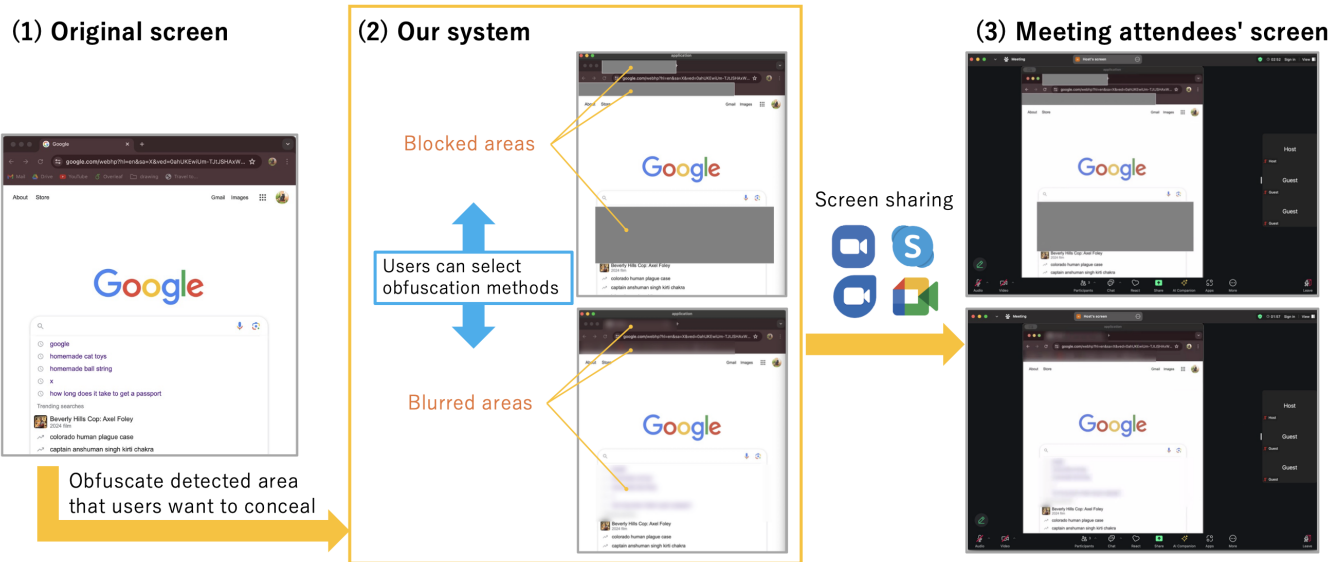


Figure 1: ScreenConcealer is a privacy-protection system for virtual meetings. (1) Original screen, (2) system automatically detects the position of web browsers and obfuscates areas where personal data and private information are detected using an object-detection model. Blocking (upper) and blurring (bottom) can be selected as obfuscation methods, (3) Meeting attendee's view (i.e., shared screen).

ABSTRACT

During screen sharing in virtual meetings, there have been many cases of unintentional disclosure of personal data that can directly identify users (e.g., names or addresses) or private information that directly does not identify users but that they do not want to share (e.g., users' preferences or online activities). We propose a system that protects personal data and private information on web browsers from being viewed by others. The system automatically takes screenshots of the web-browser window and detects personal data and private information from the images. The system then obfuscates the detected area with blocking or blurring, and displays it as a duplicated window. We present an experiment on a model that detects personal data and private information on web browsers

to implement our system. We also introduce several features to improve user experience.

CCS CONCEPTS

• **Human-centered computing** → Interaction techniques; Interactive systems and tools; • **Security and privacy** → Web application security.

KEYWORDS

Privacy, Personal data, Virtual meetings, Screen sharing

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1 INTRODUCTION

The opportunities for virtual meetings have significantly increased due to the restrictions on physical contact caused by the spread of COVID-19 [3]. Business meetings, distance learning, and personal

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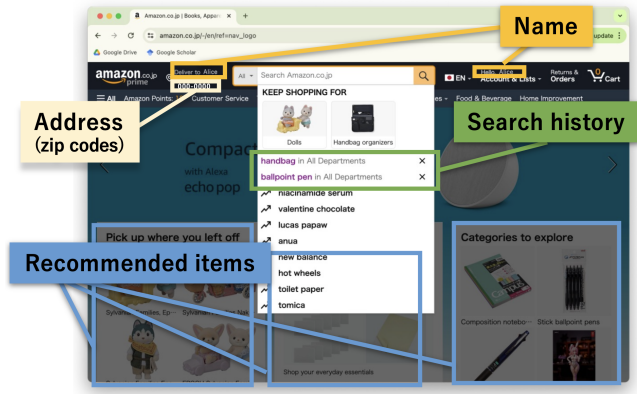


Figure 2: Example of the shared screen. User’s name, address, search history, and recommended items are disclosed.

gatherings are commonly conducted using virtual-meeting tools such as Zoom (Zoom Video Communications), Teams (Microsoft), and Google Meet (Google) [16]. These tools enable real-time communication and the ability to share visual information through screen sharing, enhancing the efficiency of meetings [12, 19]. However, the shared content can sometimes include information that presenters do not want others to see, as shown in Figure 2. If presenters share such a screen, the information becomes unintentionally disclosed to all meeting attendees. Visual content contains a large amount of important information, and Zhao et al. [18] discussed various privacy risks, concerns, and solutions for visual content. Our previous study showed that users do not want to share neither *personal data* that can directly identify them (e.g., names or addresses) nor *private information* that reveals their online activities and preferences (e.g., users’ preferences or online activities) [5]. In addition, users feel embarrassed and uncomfortable when such personal data and private information are inadvertently exposed during screen sharing [5, 8, 11, 15, 17].

Several systems have been developed to protect privacy on shared screens, primarily for face-to-face meeting situations [1, 9]. Naemura et al. [9] proposed a system that displays only the windows presenters select on external displays, while presenters can see all windows, including pop-up windows. Berry et al. [1] used accessibility application programming interfaces (APIs) and application-specific APIs to detect and visually conceal private information displayed in off-the-shelf applications. However, presenters are required to set up these systems either in advance or at the time they use them, such as defining keywords to find private information.

We propose a system that automatically obfuscates areas that users do not want to share during screen sharing. We collected screenshots from a web browser to construct an object-detection model to detect items that the user wants to conceal. We then implemented our system with this model to visually protect information in real time. Our object-detection model does not rely on information from presenters. Presenters do not have to manually register personal data and private information when they use our system, as was required in previous studies.

2 MODEL IMPLEMENTATION

When developing the system, we considered using an object-detection model to detect areas where the users want to conceal. First, we conducted a data-collection task to prepare datasets for implementing an object-detection model. Ten volunteers participated in the task (mean age of 22.3 years, SD = 1.3, six females and four males). Participants were asked to browse the five websites listed in Table 1 and complete the tasks on each website in Google Chrome (hereinafter Chrome) on their own laptop PCs (MacBooks). We mainly selected the top sites by traffic in Japan in July 2023, according to the SimilarWeb’s Top Websites Ranking¹, making sure there was no overlap in site categories. The participants were asked to complete tasks specific to the website, e.g., searching for keywords on Google Search and watching a created playlist on YouTube. Screenshots of the Chrome window were taken while participants were doing the tasks by using an application implementing AppleScript to capture the specified window as images when a click event was detected and every five seconds. Participants then annotated the collected images of themselves for the model to learn from by using labelImg². We used the 14 labels, shown in Table 1. These labels were selected on the basis of a previous study that clarified specific items users do not want to share on websites [5].

We collected 9041 images and 27195 annotated areas. One of the authors reviewed the submitted images and annotated regions to ensure that they were annotated correctly. The created datasets were divided into training and validation data using the leave-one-out method, which involves training a model on data from all but one user and validating it on the data from the remaining user. We used the YOLOv5s model from YOLOv5 [13] to train the collected datasets. YOLOv5s is both accurate and lightweight, which are necessary for our system. Training parameters included image size of 640, batch size of 32, number of epochs of 50, bounding box interval of 1, and weights of yolov5s.pt. The model performance is shown in Table 1.

3 SYSTEM OVERVIEW

We developed our system using the implemented object-detection model. An overview of our system is shown in Figure 1. The user first launches the system, being opened a web-browser window that intends to share. The system automatically detects the position of the web-browser window and captures it as an image, and the trained object-detection model stored locally on user’s PC is applied to detect the areas to be concealed. A new window is then created to display the captured image, with the area of the detected items obfuscated. The user can designate the new window as a shared window for a virtual meeting in a visually protected state from meeting attendees.

This system also includes the following features:

Selectable obfuscation methods: Blocking and blurring are commonly used methods for obfuscating images [4, 7, 14]. Blocking is superior in terms of privacy robustness. However, attendees viewing obfuscation generally prefer blurring to blocking [7, 14]. Blurring preserves visual information such as color or movement to be hidden while protecting presenter’s privacy, which contributes

¹<https://www.similarweb.com/top-websites/>

²<https://github.com/HumanSignal/labelImg>

Table 1: Results of validation of object-detection model for detecting personal data and private information.

Website	Site category	Label	Instance	Precision	Recall	mAP50	F1
Amazon	Marketplace	amazon_address	151	0.725	0.752	0.752	0.734
		amazon_name	238	0.678	0.724	0.621	0.694
		amazon_recommend	167	0.616	0.626	0.584	0.578
		amazon_searchlog	12	0.814	0.795	0.800	0.743
Chrome	Browser / Search engine	chrome_bookmark	758	0.625	0.078	0.203	0.095
		chrome_tab	902	0.990	0.993	0.988	0.991
Google Search	Search engine	google_searchlog	28	0.886	0.708	0.781	0.775
X (Twitter)	Social media networks	twitter_icon	209	0.747	0.618	0.668	0.661
		twitter_timeline	86	0.804	0.813	0.852	0.749
		twitter_recommend	120	0.928	0.798	0.928	0.810
		twitter_searchlog	17	0.834	0.841	0.865	0.821
Yahoo! News	News & media publishers	yahoo_advertisement	156	0.778	0.823	0.826	0.787
YouTube	Streaming & online TV	youtube_searchlog	20	0.708	0.827	0.785	0.742
		youtube_recommend	134	0.873	0.765	0.809	0.780
All of the above			2,720 (sum)	0.788 (avg.)	0.728 (avg.)	0.745 (avg.)	0.755 (avg.)

to high users' satisfaction [2, 7]. We implemented blocking and blurring as obfuscation methods for presenters to select as shown in Figure 1.

Partial revealing for over-obfuscated areas: Our system may obfuscate areas that the user actually wants to share. To address this, we implemented a feature that enables users to reveal the content of an over-obfuscated area by clicking on it. When users click, the obfuscation is revealed, making the content visible.

Obfuscation levels: Among the labels used to implement the model, the degree to which users do not want to share varies [5]. For instance, 97.3% of participants indicated that they did not want to share their home address displayed on Amazon, while 48.6% did not want to share their name displayed on Amazon. Users also disclose their information differently depending on their relationships because privacy is closely related to trust [6, 10]. Accordingly, we prepared three levels of obfuscation: "Low" obfuscates information that over 80% of participants did not want to share, "Medium" obfuscates information that over 60% did not want to share, and "High" obfuscates all detected information. Therefore, users can select an appropriate obfuscation level on the basis of their preferences and meeting situations.

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